

CSS Margins

- The CSS margin properties are used to generate space around elements.
- Units : px,pt,cm,%
- Margin-This property is used to set all the properties in one declaration.
- margin-left /margin-right/margin-top/margin-bottom

CSS has properties for specifying the margin for each side of an element:

- [margin-top](#) - sets the top margin of an element
- [margin-right](#) - sets the right margin of an element
- [margin-bottom](#) - sets the bottom margin of an element

- [margin-left](#) - sets the left margin of an element
- auto - the browser calculates the margin
- *length* - specifies a margin in px, pt, cm, etc.
- % - specifies a margin in % of the width of the containing element
- inherit - specifies that the margin should be inherited from the parent element

<!DOCTYPE html>

<html>

<head>

<style>

```
div {  
  border: 1px solid black;  
  margin-top: 100px;  
  margin-bottom: 100px;  
  margin-right: 150px;  
  margin-left: 80px;  
  background-color: lightblue;  
}  
</style>  
</head>  
<body>  
<h2>Using individual margin properties</h2>  
<div>This div element has a top margin of  
100px, a right margin of 150px, a bottom  
margin of 100px, and a left margin of  
80px.</div>  
</body>
```

</html>

Shorthand Property

[margin-top](#) [margin-right](#) [margin-bottom](#)
[margin-left](#)

margin: 25px 50px 75px 100px;

<!DOCTYPE html>

<html>

<head>

<style>

div {

border: 1px solid black;

margin: 25px 50px 75px 100px;

background-color: lightblue;

}

</style>

</head>

<body>

<h2>The margin shorthand property - 4 values</h2>

<div>This div element has a top margin of 25px, a right margin of 50px, a bottom margin of 75px, and a left margin of 100px.</div>

<hr>

</body>

</html>

Negative values are allowed. Default value:0

All the margin properties can have the following values:

- . auto - the browser calculates the margin and horizontally center the element and the remaining space will be split equally between the left and right margins**

- ***length*** - specifies a margin in px, pt, cm, etc.

If the [margin](#) property has one value:

- **margin: 25px;**
 - all four margins are 25px

margin: 25px 50px;

- top and bottom margins are 25px
- right and left margins are 50px

The inherit Value

You can set the [margin](#) property to inherit to let the margin be inherited from the parent element.

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<style>
```

```
div {
```

```
    border: 1px solid red;
```

```
    margin-left: 100px;
```

```
}
```

```
p.ex1 {
```

```
    margin-left: inherit;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h2>Use of the inherit value</h2>
```

```
<p>Let the left margin be inherited  
from the parent element:</p>
```

```
<div>
```

```
<p class="ex1">This paragraph has an  
inherited left margin (from the div  
element).</p>
```

```
</div>
```

```
</body>
```

```
</html>
```

CSS Padding

CSS Padding property is used to define the space between the element content and the element border.

- `div {
padding-top: 50px;
padding-right: 30px;
padding-bottom: 50px;
padding-left: 80px;
}`

- OR

- `div {
padding: 25px 50px 75px 100px;
}`

padding shorthand property-

padding: 25px 50px 75px 100px;

- top padding is 25px
- right padding is 50px
- bottom padding is 75px
- left padding is 100px

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<style>
```

```
div {
```

```
    border: 1px solid black;
```

```
    padding: 25px 50px 75px;
```

```
    background-color: lightblue;
```

```
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h2>The padding shorthand property -  
3 values</h2>
```

```
<div>This div element has a top  
padding of 25px, a right and left  
padding of 50px, and a bottom padding  
of 75px.</div>
```

```
</body>
```

```
</html>
```

Assignment

Create a `<div>` box.

Set:

- Top margin: 20px
- Right margin: 30px

- . Bottom margin: 20px
- . Left margin: 30px

Add background color lightblue.

Add 10px padding inside the box.

The [min-width](#) and [max-width](#) properties are used to set the minimum and maximum width of an element.

The [min-height](#) and [max-height](#) properties are used to set the

minimum and maximum height of an element.

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
    <title>Min and Max Width
```

```
Example</title>
```

```
    <style>
```

```
        .box {
```

```
            background-color: lightgreen;
```

```
            border: 2px solid black;
```

```
            padding: 15px;
```

```
            min-width: 200px;
```

```
            max-width: 500px;
```

```
        }
```

```
    </style>
```

```
</head>
```

```
<body>
```

```
  <div class="box">
```

This box has min-width of 200px
and max-width of 500px.

Try resizing the browser window
to see the effect.

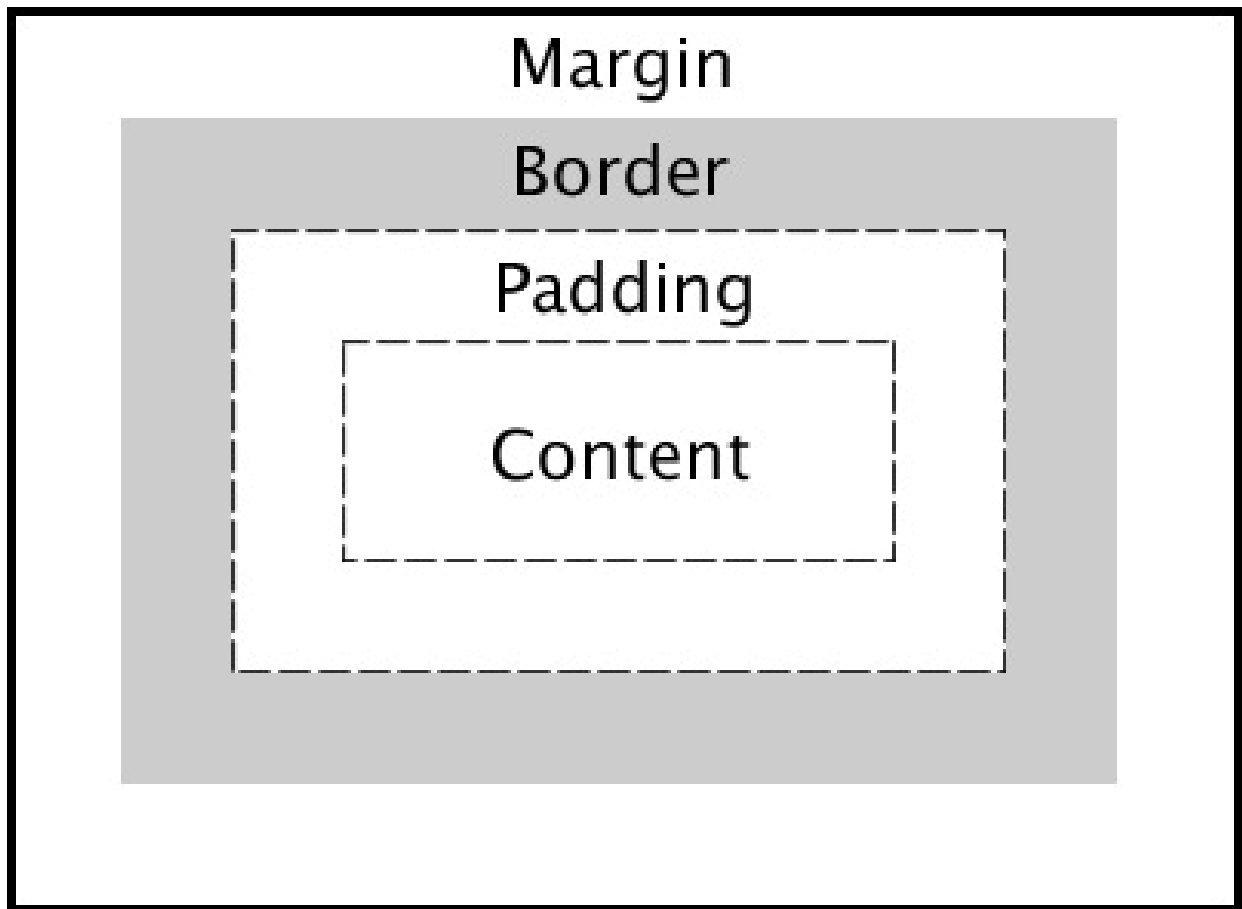
```
  </div>
```

```
</body>
```

```
</html>
```

What is CSS Box Model?

Every HTML element is treated as a
rectangular box in CSS.



The box has 4 parts:

Parts of Box Model

1. Content

- Actual text or image inside the element.

- . Controlled by width and height.

2. Padding

- . Space **inside** the element.
- . Between content and border.
- . The padding is transparent

3. Border

- . Surrounds padding and content.

4. Margin

- . Space **outside** the element.
- . Creates distance between elements.
- . The margin is transparent

The total width of an element should be calculated like this:

Total element width = width + left padding + right padding + left border + right border

The total height of an element should be calculated like this:

Total element height = height + top padding + bottom padding + top border + bottom border

assignment

Write HTML + CSS code to:

- . Create a div
- . Width: 250px

- . Height: 120px
- . Padding: 15px
- . Border: 4px solid red
- . Margin: 20px
- . Background color: lightyellow

Soln-

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Box Model Test</title>
<style>
  .box {
    width: 250px;
    height: 120px;
    padding: 15px;
    border: 4px solid red;
    margin: 20px;
    background-color: lightyellow;
  }
</style>
</head>
<body>

  <div class="box">
```

Box Model Assessment

</div>

</body>

</html>

An outline is a line that is drawn around elements, OUTSIDE the borders, to make the element "stand out"

<!DOCTYPE html>

<html>

```
<head>
```

```
<style>
```

```
p {
```

```
outline-color:green;
```

```
padding: 5px;
```

```
}
```

```
p.dotted {outline-style: dotted;}
```

```
p.dashed {outline-style: dashed;}
```

```
p.solid {outline-style: solid;}
```

```
p.double {outline-style: double;}
```

```
p.groove {outline-style: groove;}
```

```
p.ridge {outline-style: ridge;}
```

```
p.inset {outline-style: inset;}
```

```
p.outset {outline-style: outset;}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h2>The outline-style Property</h2>
```

```
<p class="dotted">A dotted  
outline</p>
```

```
<p class="dashed">A dashed  
outline</p>
```

```
<p class="solid">A solid outline</p>
```

```
<p class="double">A double  
outline</p>
```

```
<p class="groove">A groove outline.  
The effect depends on the outline-color  
value.</p>
```

<p class="ridge">A ridge outline. The effect depends on the outline-color value.</p>

<p class="inset">An inset outline. The effect depends on the outline-color value.</p>

<p class="outset">An outset outline. The effect depends on the outline-color value.</p>

</body>

</html>

Margin

Space outside
element

Padding

Space inside
element

Margin

Separates
elements

Transparent

Padding

Creates space inside
border

Takes background
color

Remember margin is outside the
border.

Only padding and content area show
background color.